

M02 - Data Versioning and Lineage



Topic

- **Why Reproducibility Matters**
- **Dataset Versioning**
- **Metadata Management**
- **Experiment Tracking**
- **Data & Model Lineage**
- **Data Governance**
- **Hands-on with MLflow & DVC**
- **Module Summary & Key Takeaways**





Why Reproducibility Matters

Why Reproducibility Matters





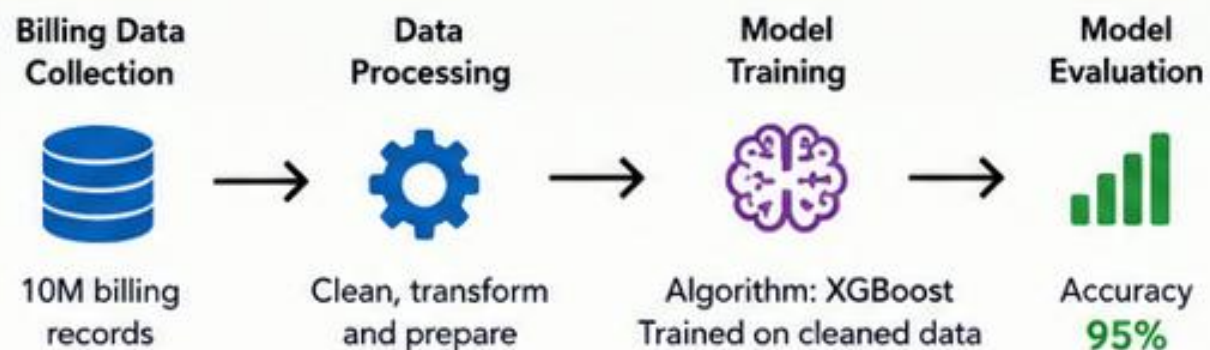
Scenario:

Suppose Vodafone trains an AI model to detect billing anomalies and achieves **95% accuracy**.



Six months later, the team wants to recreate that same model. If they don't know which dataset was used, which algorithm was selected, what parameters were applied, or which version of the code was used, they may get a **completely different result**.

1. INITIAL MODEL TRAINING (Month 0)

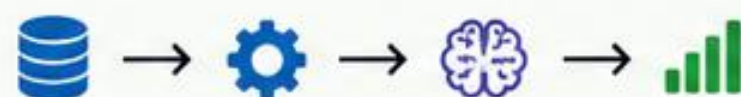


WHAT WAS USED (Known and Tracked)

	Dataset	billing_data_v1.2.csv (Snapshot on 01-Jan-2025)
	Algorithm	XGBoost
	Parameters	n_estimators=200, max_depth=6, learning_rate=0.05, subsample=0.8
	Code Version	Git Commit: a1b2c3d (branch: main)
	Environment	Python 3.10, XGBoost 1.7.6, Pandas 2.1.0
	Result	Model Accuracy: 95%

2. RECREATING THE MODEL AFTER 6 MONTHS (Month 6)

A. IF EVERYTHING IS KNOWN AND TRACKED



- ✓ Use the same dataset version (billing_data_v1.2.csv)
- ✓ Use the same algorithm (XGBoost)
- ✓ Use the same parameters
- ✓ Use the same code version (a1b2c3d)
- ✓ Same environment

RECREATED MODEL RESULT



Same Accuracy Achieved!
Model is Reproducible



B. IF INFORMATION IS MISSING OR DIFFERENT



- ✗ Different dataset (new or unknown version)
- ✗ Different algorithm or version
- ✗ Different or unknown parameters
- ✗ Different code version
- ✗ Different environment / library versions

RECREATED MODEL RESULT



Lower Accuracy
Completely Different Result!



REPRODUCIBILITY DELIVERS



Reliable
Results



Easier Debugging
& Troubleshooting



Audit & Compliance
Ready



Build Trust
in AI



Consistent Performance
Over Time

REPRODUCIBILITY DELIVERS



**Reliable
Results**



**Easier Debugging
& Troubleshooting**



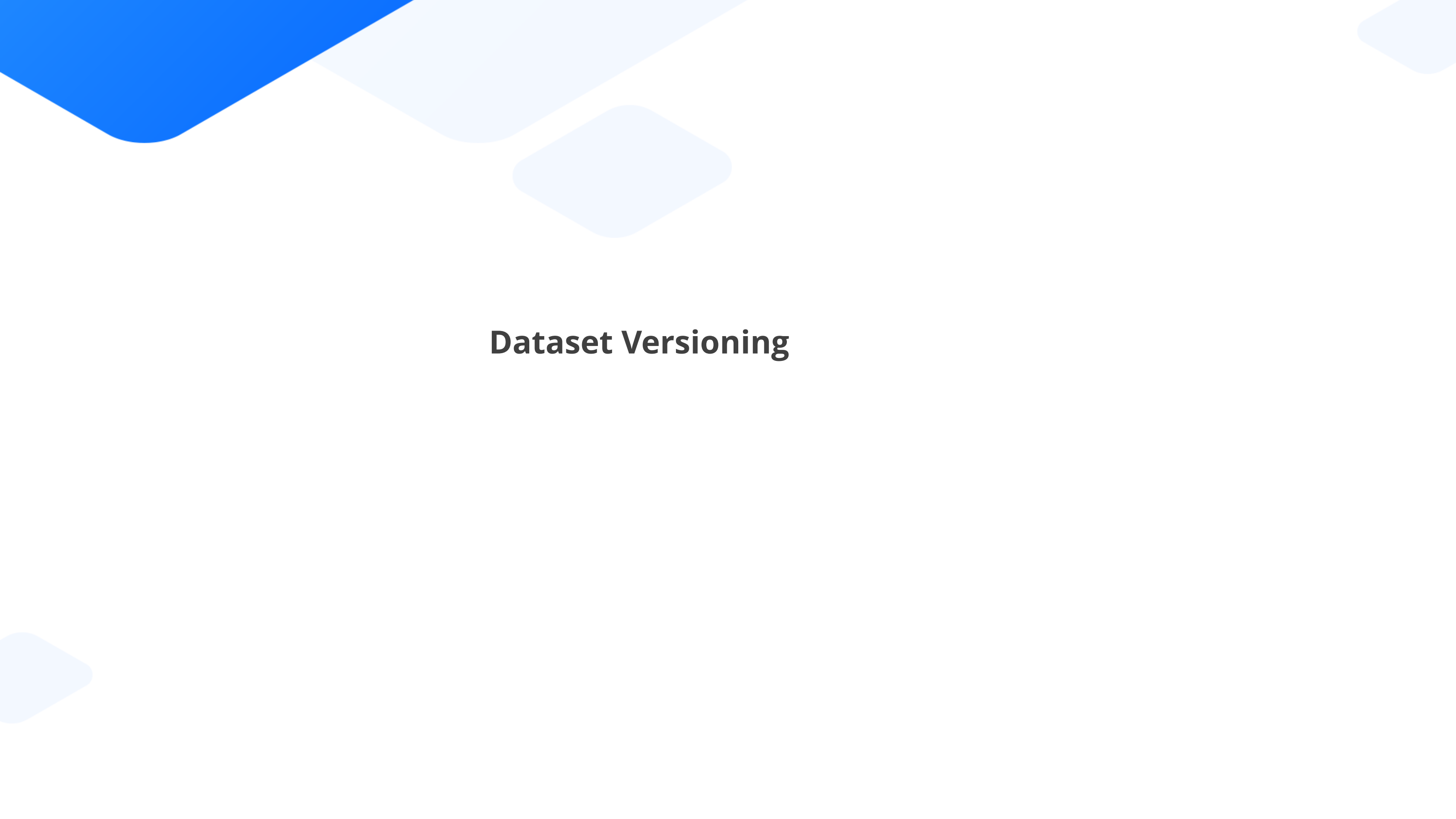
**Audit & Compliance
Ready**



**Build Trust
in AI**

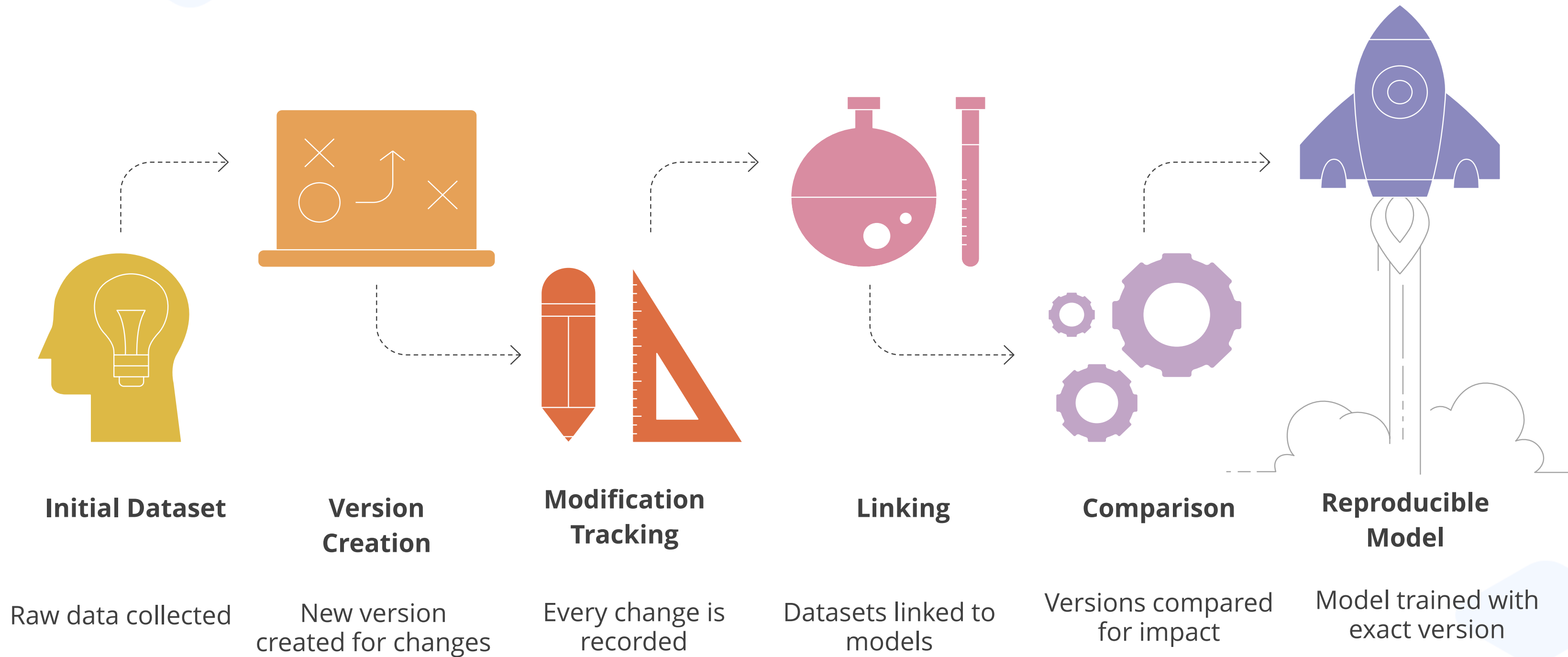


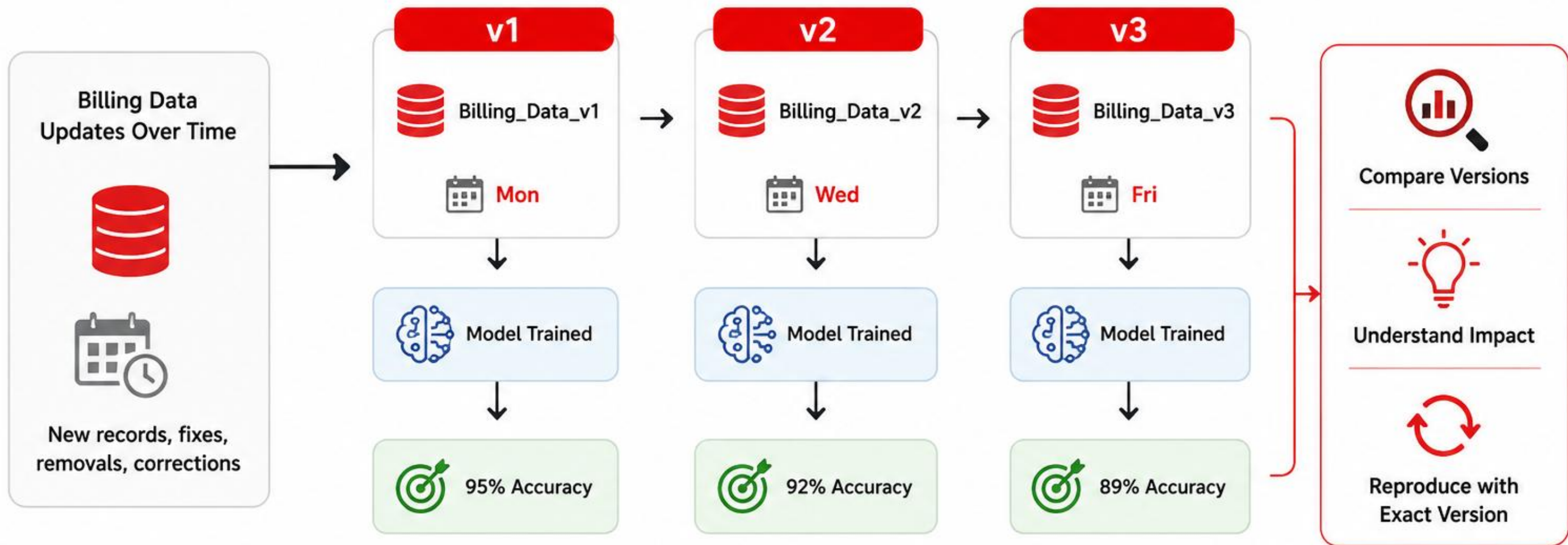
**Consistent Performance
Over Time**



Dataset Versioning

Dataset Versioning Process





WHY IT MATTERS



Need to Recreate
or Investigate?



Select Exact Dataset
Version (e.g., v1)



Retrain Model



Reproduce Same
Result (95%)

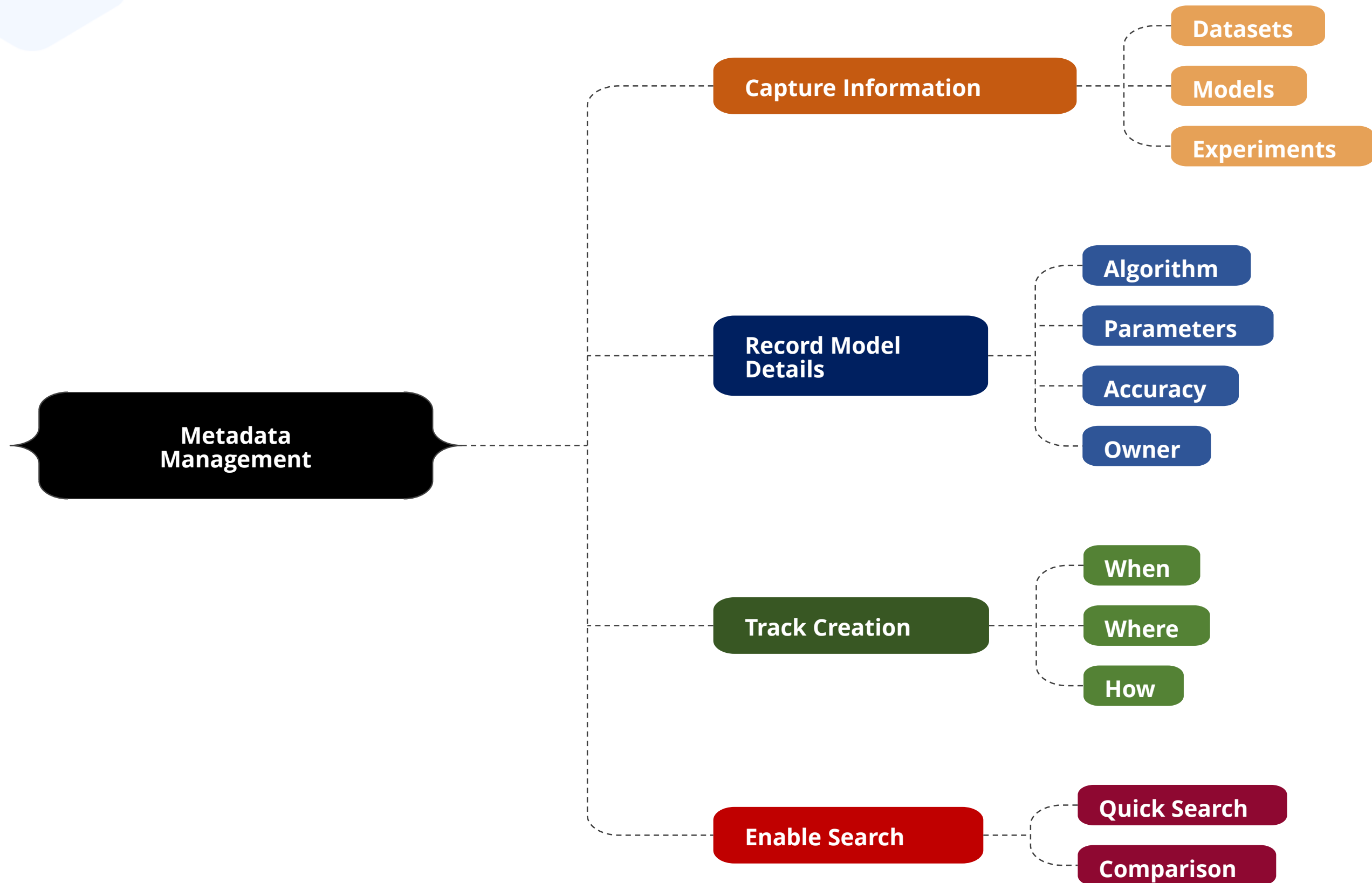


Reproducible
Reliable
Trusted



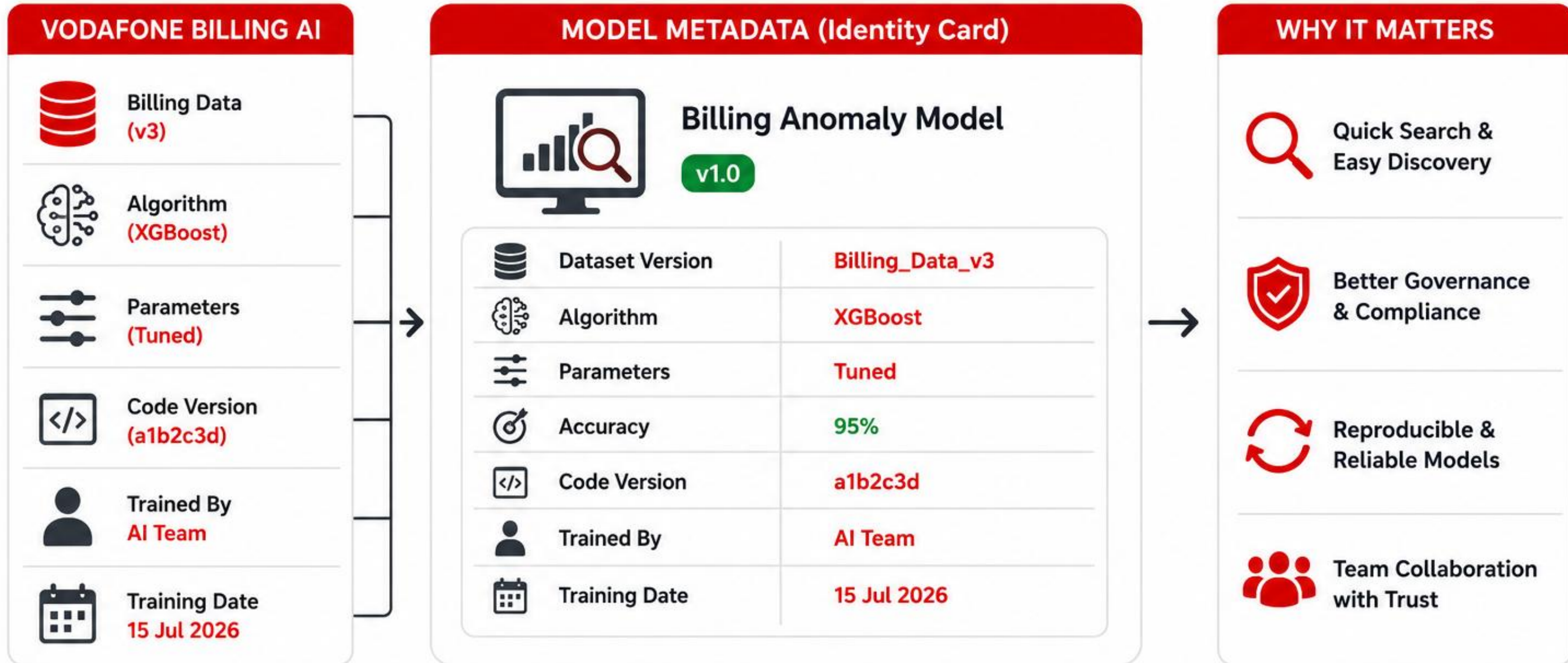
Metadata Management

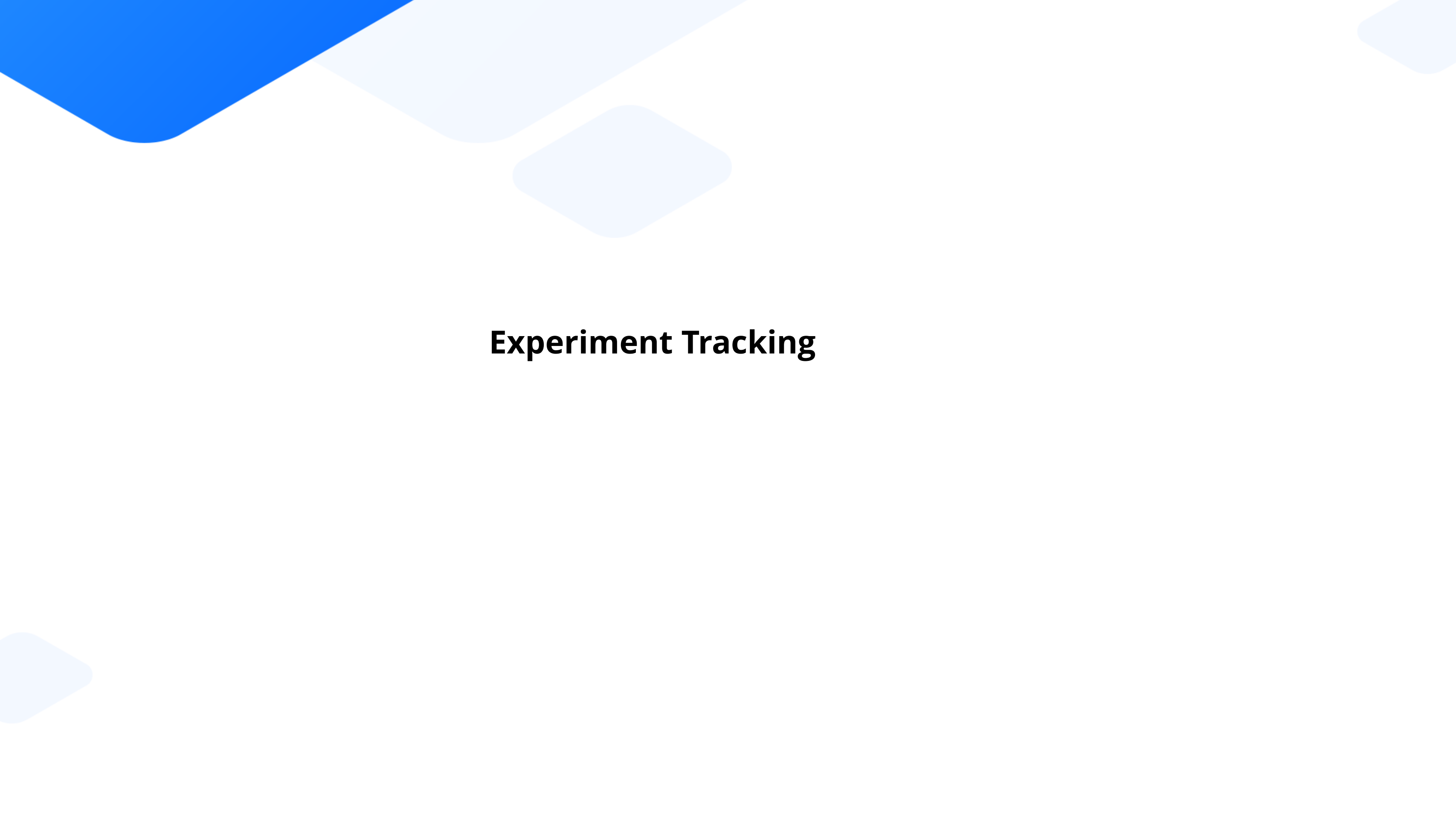
Metadata Management



METADATA MANAGEMENT – VODAFONE BILLING AI

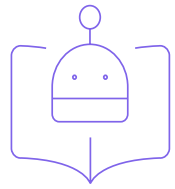
Every model has its identity. Information is easy to find, trust and reuse.





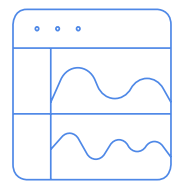
Experiment Tracking

Experiment Tracking



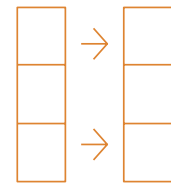
Record every training experiment

- Capture all parameters and configuration settings



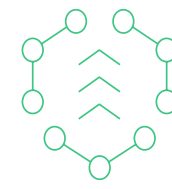
Track performance metrics

- Monitor accuracy precision recall and F1-score



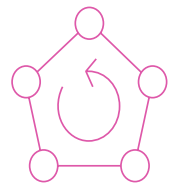
Compare multiple model runs

- Identify the best-performing model



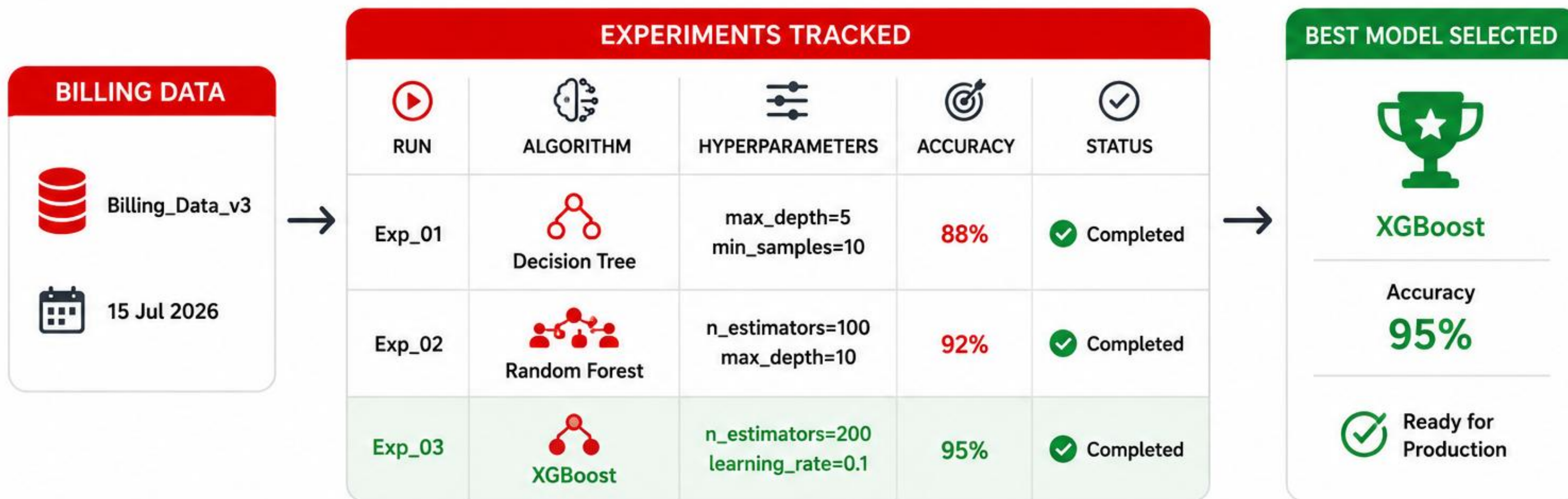
Log datasets, code versions, and hyperparameters

- Document inputs used in each experiment



Enable reproducibility

- Recreate any experiment whenever needed





Data & Model Lineage

Terminology

What is Lineage?

Lineage means **the complete history or journey of something**.

In MLOps, **Data & Model Lineage** answers questions like:

- **Where did this data come from?**
- **How was it transformed?**
- **Which features were created?**
- **Which dataset trained the model?**
- **Which model generated this prediction?**

Think of lineage as a **family tree** or a **Google Maps route**. It shows the complete path from the **source of the data** all the way to the **final AI prediction**.

Understand Data & Model Journey

Data Source

Initial data collection

Model Training

Building and refining models

Model Deployment

Making models available for use

Data Processing

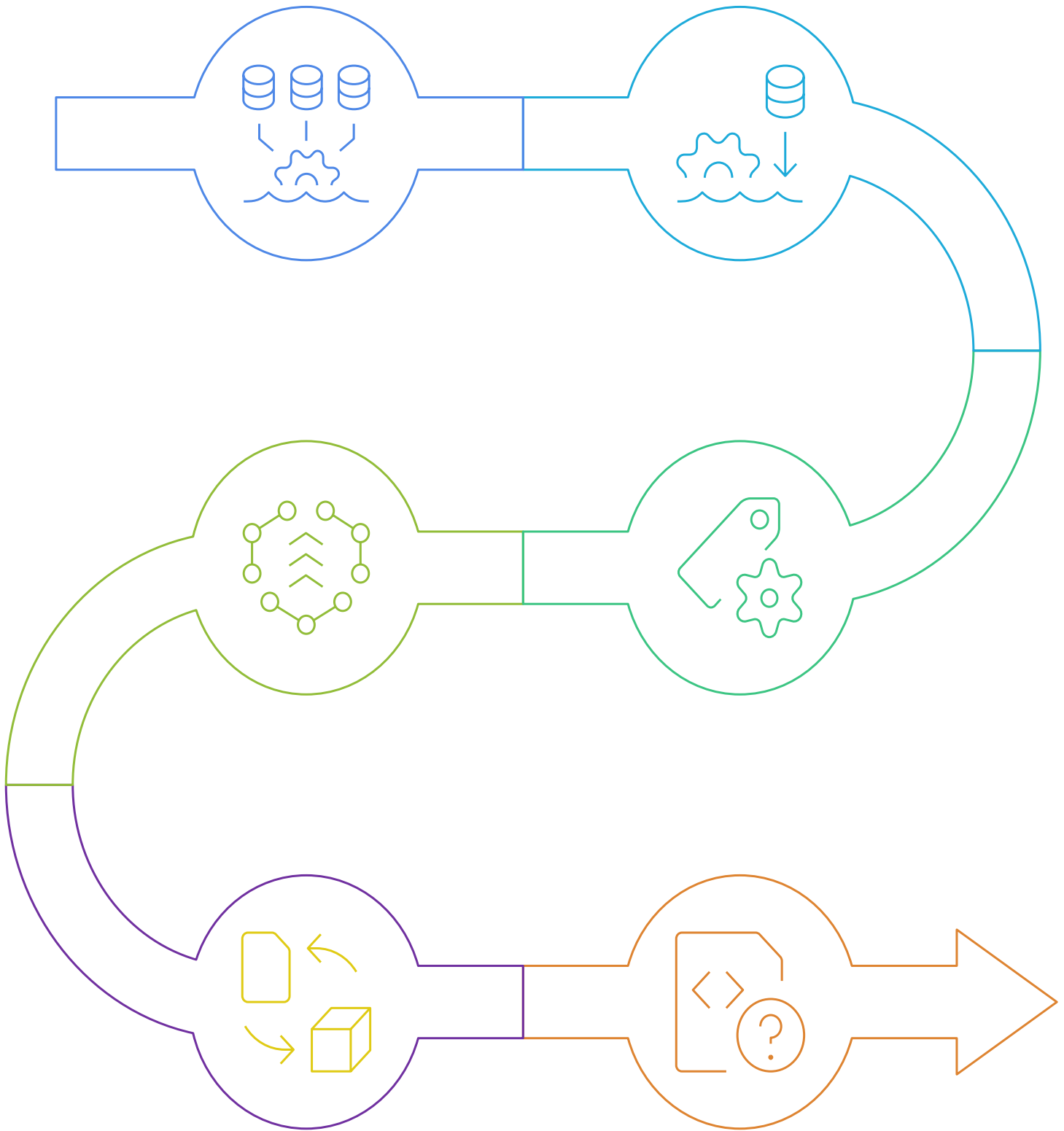
Data cleaning and transformation

Feature Engineering

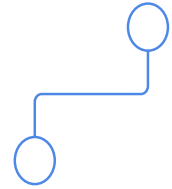
Creating relevant features

Predictions

Generating predictions from models



Data & Model Lineage



**Trace data
journey**

Trace the complete journey of data from source to deployed model.



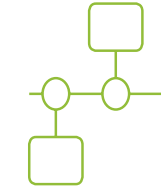
**Track
relationships**

Track relationships between datasets, features, models, and predictions.



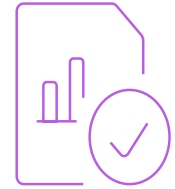
Identify origin

Identify the origin of every model and its training data.



**Simplify
impact
analysis**

Simplify impact analysis when data or models change.



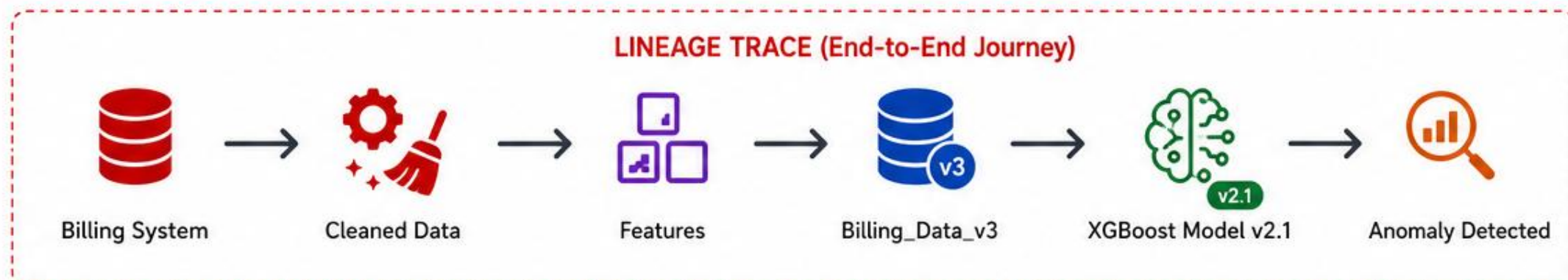
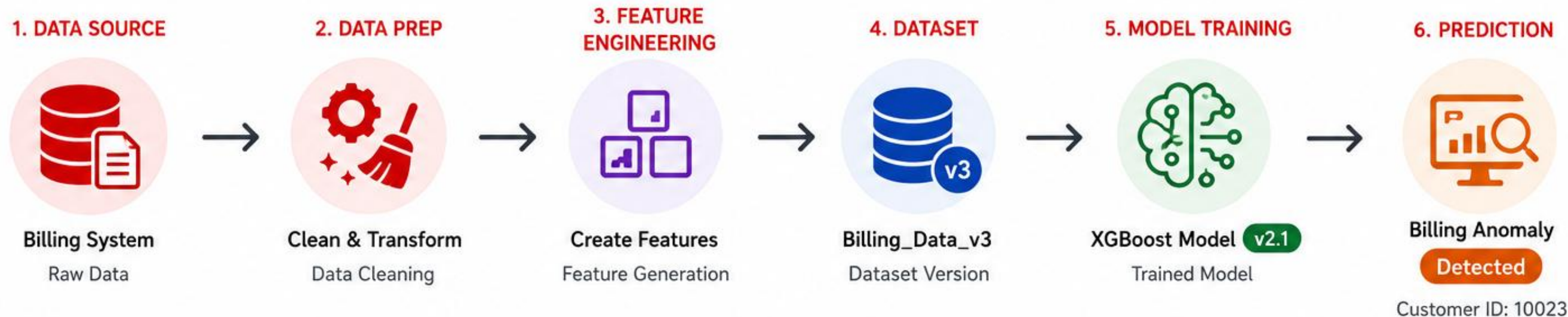
**Support
governance
and auditing**

Support governance, auditing, and reproducibility with end-to-end traceability.



DATA & MODEL LINEAGE – VODAFONE BILLING AI

From Source to Prediction



WHY IT MATTERS



Find Source
of Any Issue



Understand Impact
of Changes



Audit &
Compliance



Reproducibility
& Trust



Build Confidence
in AI

EXAMPLE



Customer asks:
"Why was my bill
flagged as anomaly?"



Lineage shows the
complete path



Root cause found
faster



More Transparency,
More Trust



Dataset Version
Billing_Data_v3



Model Version
XGBoost v2.1



Training Date
15 Jul 2026



Model Accuracy
95%



Trained By
AI Team



Data Governance

Data Governance



Define policies

**Secure and
responsible use of
data**



Control access

**Datasets, models,
and ML resources**



**Ensure data
quality**

**Validation and
standardization**



**Meet regula-
tory and
compliance**

Proper auditing



**Protect
sensitive
information**

**Enable trusted AI
development**

1. POLICIES & RULES



Define rules for data use, security and compliance.

2. ACCESS CONTROL



Right people access the right data at the right time.

3. DATA QUALITY



Ensure accurate, consistent and trusted data.

4. AUDIT & COMPLIANCE



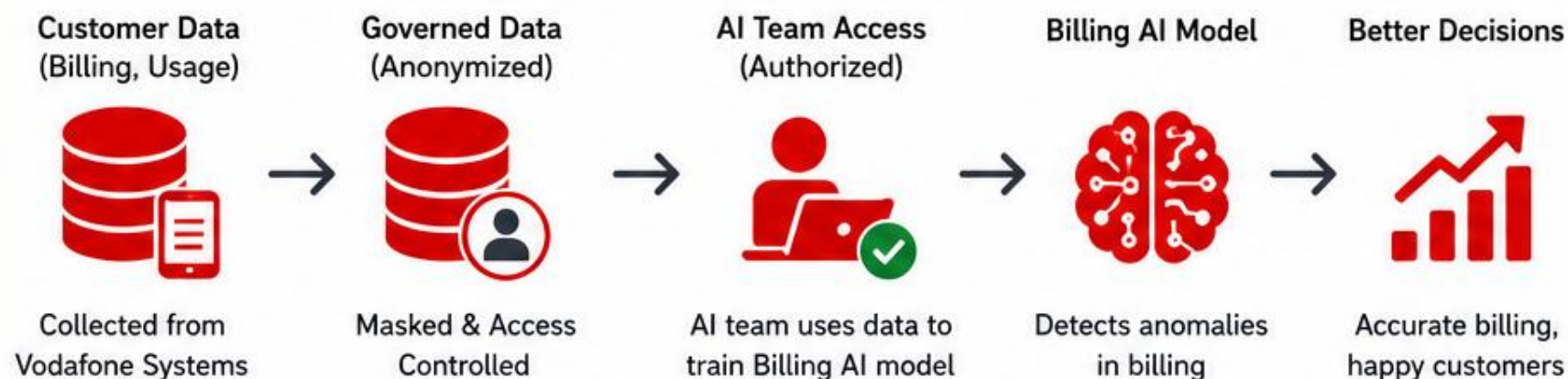
Track all access and changes for audit and compliance.

5. DATA PRIVACY



Protect customer data and meet regulatory requirements.

SIMPLE EXAMPLE – VODAFONE BILLING AI



WHO CAN ACCESS WHAT?

	Customer Support	View customer billing info	✓
	Data Scientist / AI Team	Use anonymized data to build models	✓
	System Admin	Manage systems & permissions	✓
	Unauthorized User	No access to sensitive data	✗

KEY BENEFITS



Secure Data Usage



Regulatory Compliance



High Data Quality



Trusted AI Outcomes



Customer Trust

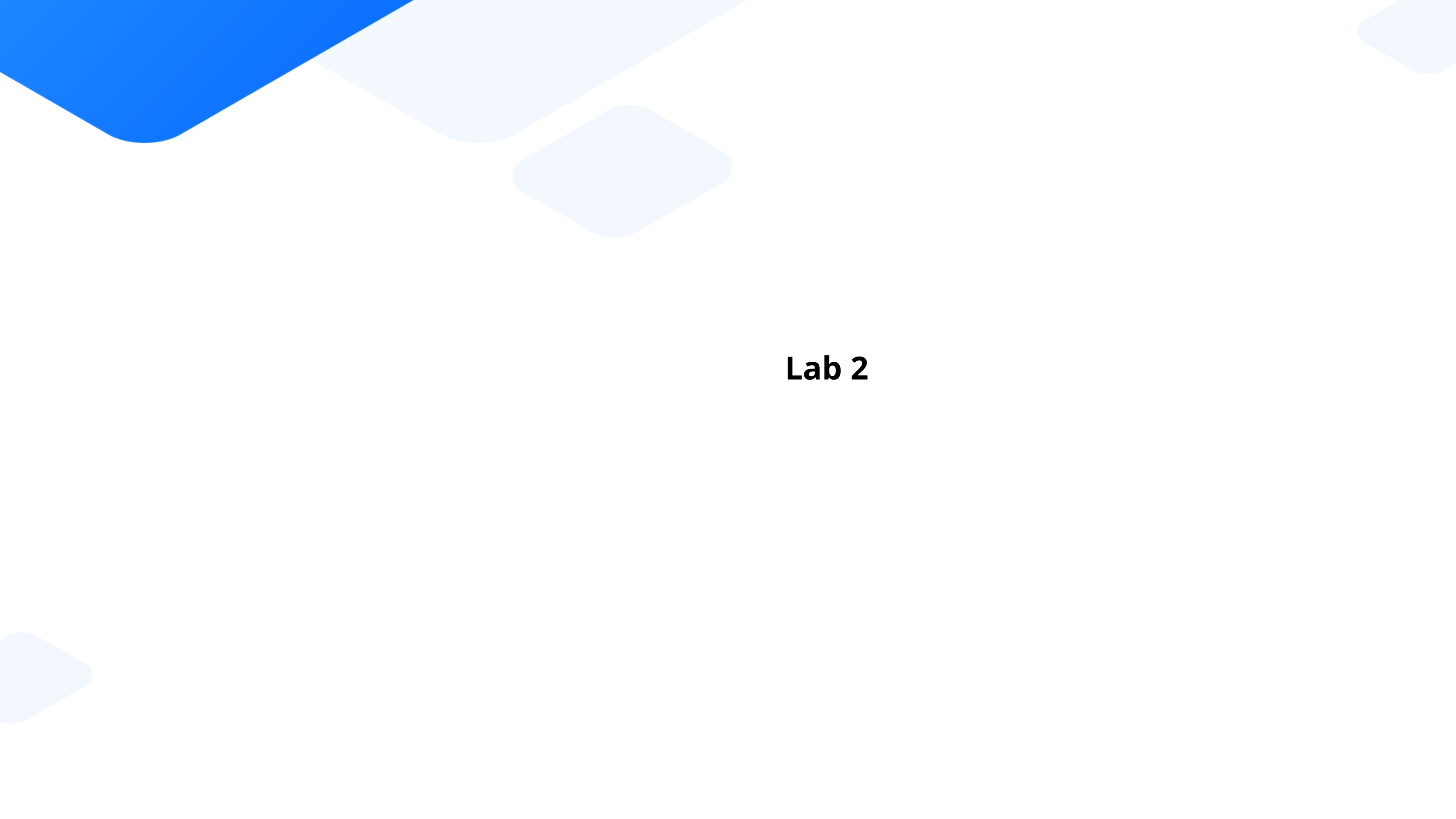


Data Governance ensures Vodafone uses data responsibly to build AI that customers can trust.

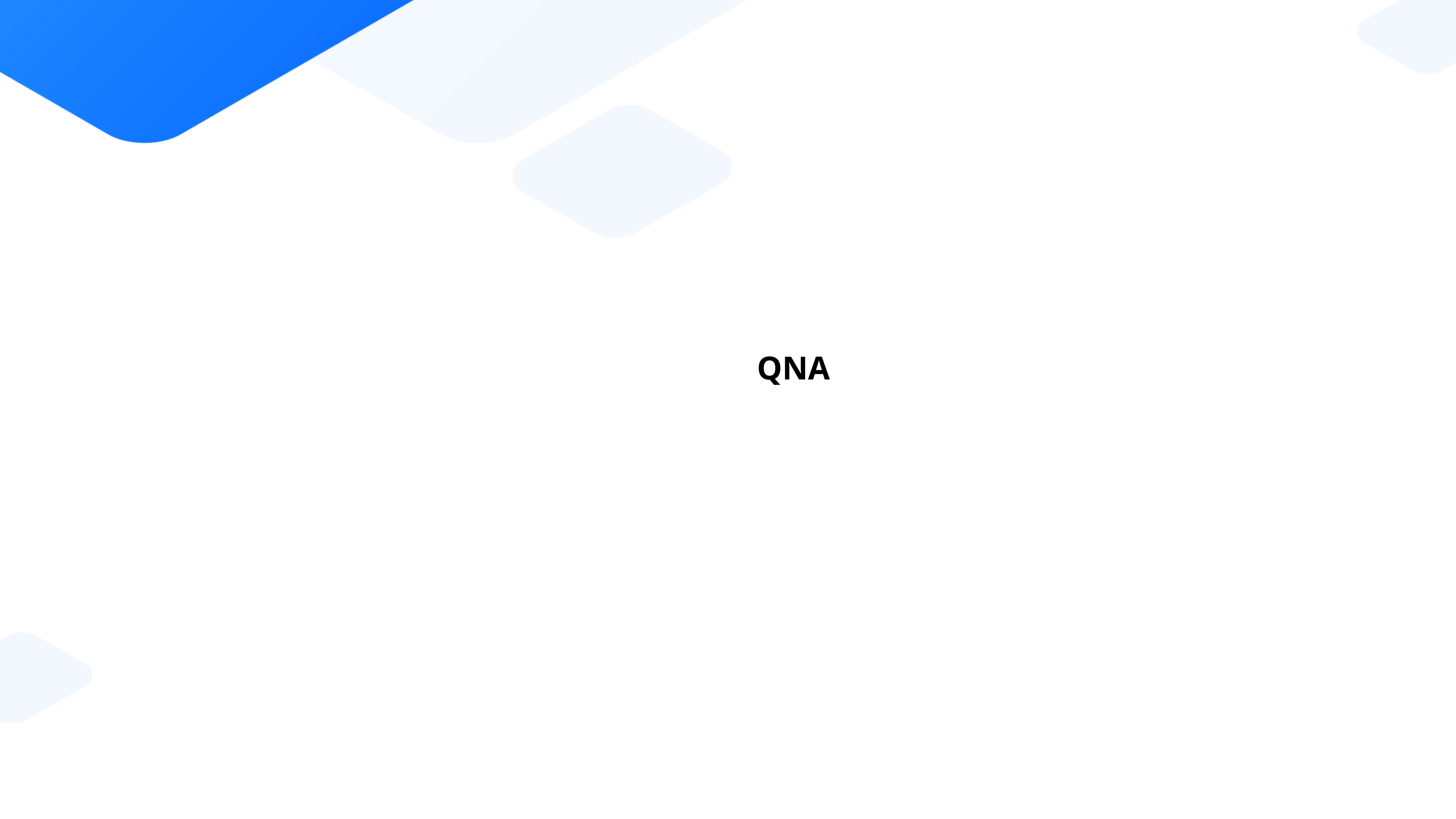


Governed Data. Trusted AI. Better Customer Experience.





Lab 2



QNA